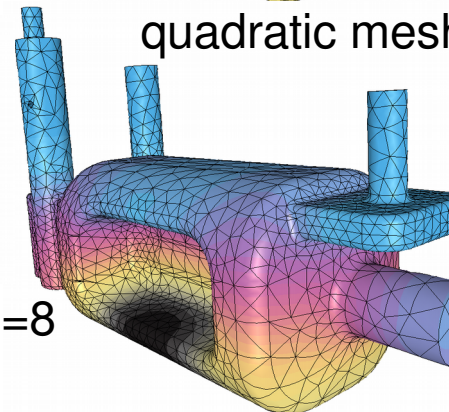
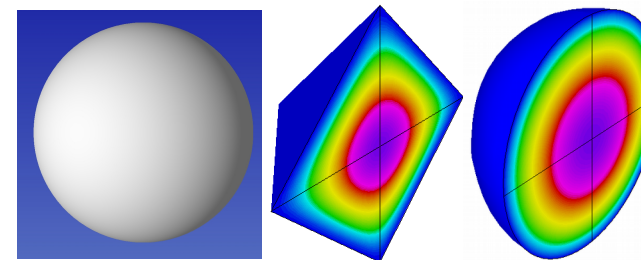
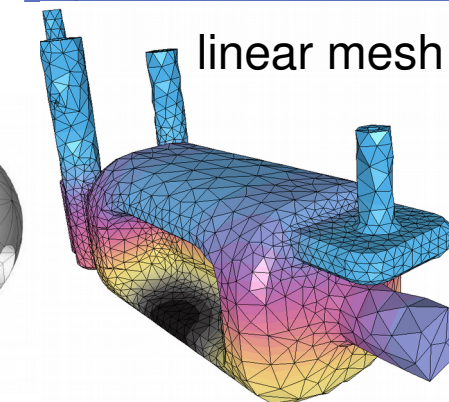
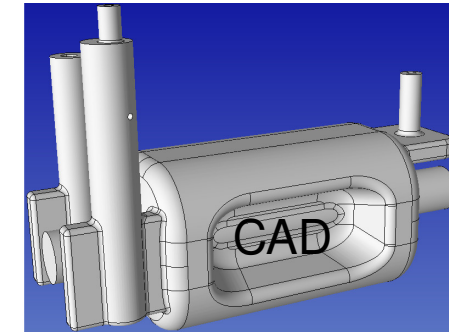
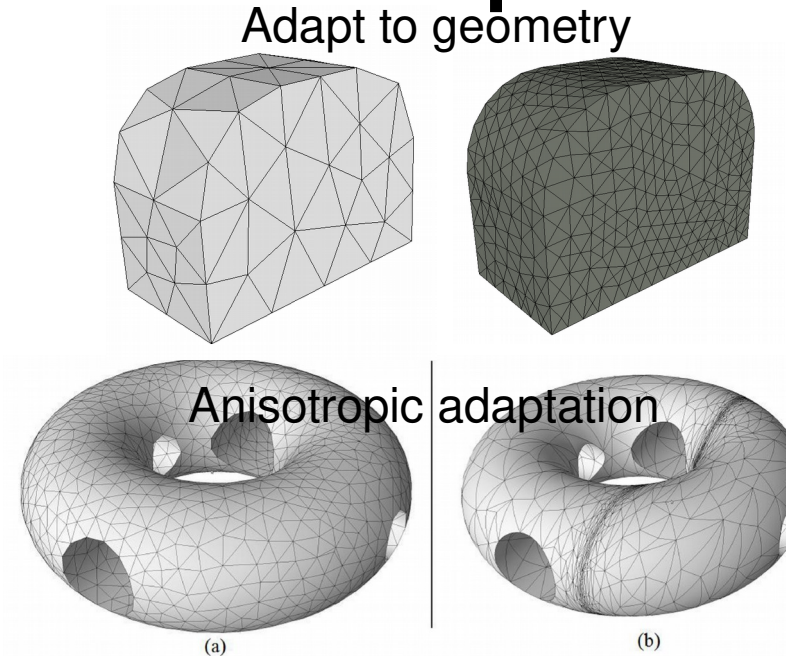


Integration of Geometry-Based Mesh Adaptation

- ❑ Parallel Unstructured Mesh Infrastructure (PUMI)
 - ❑ Supports operations on distributed meshes
 - ❑ Maintains linkage to original geometry
 - ❑ Supports conforming mesh adaptation based on anisotropic mesh adaptation based on metric field
- ❑ Status of implementation
 - ❑ Integration, including loading of parallel meshes
 - ❑ Linked to linear and quadratic element mesh generation
 - ❑ Geometric inflation to up to 6th order
 - ❑ Links to geometry can be used at any point in a workflow
 - ❑ Simple test cases executed – showing importance of geometry approximation and linkage with CAD
 - ❑ Improving curved mesh adaptation



Influence of geometry: For $P=8$
linear geometry = 0.1757
quartic geometry = 0.4156
exact value = 0.4166